AI Resume Analyzer and Job Recommendation System

Objective
Develop an AI-based application that analyzes a resume, compares it with predefined job roles, recommends suitable jobs, identifies missing skills, and generates a learning roadmap.

Technologies Used
Python, Google Colab, Pandas, NumPy, Scikit-learn, TF-IDF, Cosine Similarity, PyPDF, python-docx.

Workflow
1. Create job roles dataset.
2. Create skill dictionary.
3. Read and clean resume text.
4. Extract skills.
5. Apply TF-IDF vectorization.
6. Calculate cosine similarity.
7. Recommend the best job role.
8. Identify missing skills.
9. Generate a learning roadmap.

Result
The system successfully analyzes a resume, calculates the similarity score with predefined job roles, recommends the most suitable role, identifies missing skills, and generates a personalized learning roadmap.

Conclusion
This project demonstrates the use of Natural Language Processing techniques to assist students in evaluating resumes and identifying skills required for their target careers.